

Catalan Numbers

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1 Introduction

The Catalan numbers are a sequence of natural numbers that appear frequently in counting problems. The first n Catalan numbers for $n = 0, 1, 2, 3, \dots$ are $1, 1, 2, 5, 14, 42, 132, 429, \dots$. Upon first glance, there doesn't seem to be a pattern relating consecutive numbers, but they can be expressed recursively and with an explicit formula! We will start by talking about the problems these numbers appear in, how they were solved, and a solution for the general formula to derive the n^{th} Catalan number.

2 Applications in Problems

The n^{th} Catalan number is often denoted with C_n which is what we will be using in this article. These problems all ultimately satisfy Segner's Recurrence Formula:

$$E_n = E_2 E_{n-1} + E_3 E_{n-2} + \dots + E_{n-1} E_2$$

In terms of the Catalan numbers, this formula is often rewritten as

$$C_{n-1} = C_0 C_n + C_1 + C_{n-1} + \dots + C_n C_0 = \sum_{k=0}^n C_k C_{n-k}, C_0 = 1$$

2.1 Catalan's Problem

How many different ways can a product of n different factors be calculated by pairs if their order is prescribed? Let's understand this problem through an example where we have 4 factors x_1, x_2, x_3, x_4 . There are 5 different ordered multiplications, namely $((x_1 \cdot x_2) \cdot x_3) \cdot x_4, (x_1 \cdot (x_2 \cdot x_3)) \cdot x_4, (x_1 \cdot x_2) \cdot (x_3 \cdot x_4), x_1 \cdot ((x_2 \cdot x_3) \cdot x_4), x_1 \cdot (x_2 \cdot (x_3 \cdot x_4))$

Catalan came up with a solution and named it after himself as mathematicians liked to do.

$$C_{n-1} = \frac{(4n-6)!!!!}{n!}$$

The four exclamation marks mean that $n!!!! = n(n-4)(n-8) \dots$. This idea is equivalent to the parenthesis problem that we will go over later. To connect this to Segner's Recurrence Formula, we can think about having $n+1$ total numbers and n total operations. Let M_n be the number of multiplications of n numbers. The first i pairs can be grouped in M_i ways, and the remaining $n-i-1$ pairs in M_{n-i-1} so for each i , there are $M_i M_{n-i-1}$ ways for the situation to happen. Thus, the total would be

$$M_n = M_1 M_{n-1} + M_2 M_{n-2} + \dots + M_{n-1} M_1$$

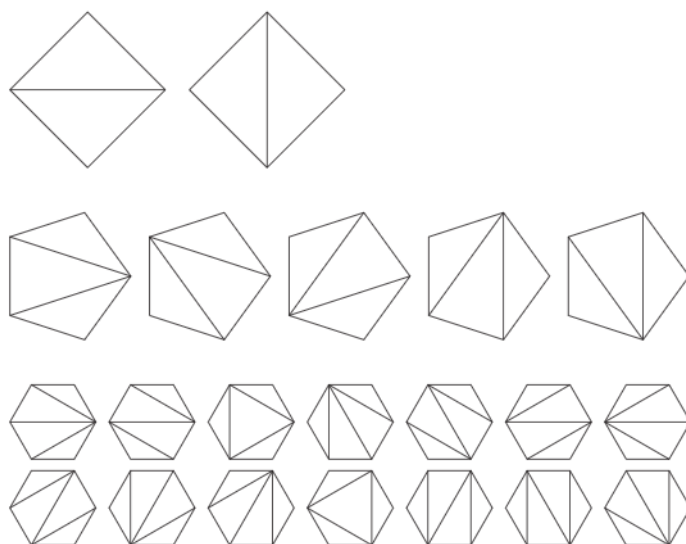
which is the same as the formula for C_{n-1}

2.2 Euler's Polygon Division Problem

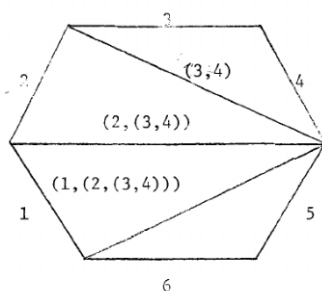
Segner's Recurrence Formula was used to solve this problem! In 1751, Euler asked how many ways a convex polygon with n sides can be divided into $n - 2$ triangles using their diagonals. Symmetry, rotations, and whether the triangles are congruent or not don't matter. The formula used was

$$E_n = \frac{2(n-2)!}{(n-1)!(n-2)!}$$

Here are the solutions to the first few polygons.



Notice that at least two of the triangles have a diagonal and two sides of the polygon as its sides. We can choose one of these triangles and remove the two sides of the polygon to break the case down into the same problem but with $n - 1$ sides. In the hexagon below, we first remove sides 3 and 4. Instead of thinking about the hexagon, we have reduced to thinking about a pentagon with the diagonal, denoted with $(3, 4)$ becoming a side. Then we remove side 2 and side $(3, 4)$. The final division is $((1, (2, (3, 4))), 5), 6$. This shares some similarities with Catalan's problem!



Section 2.6 will explain another way to think about this problem using binary trees.

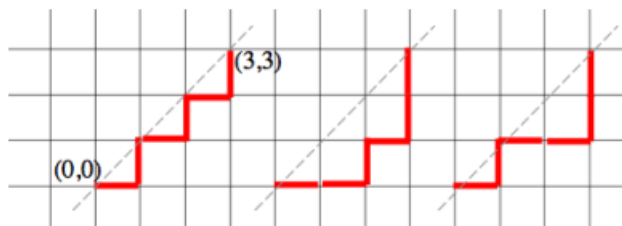
2.3 Dyck Paths

A Dyck Path is a series of up and down steps. The path goes from left to right. It begins and ends at the same level, and does not go below the starting point at any point. What is the total number of Dyck Paths of length $2n$? Once again, Catalan numbers begin to show up

$n = 0$:	*	1 way
$n = 1$:	$\wedge \backslash$	1 way
$n = 2$:	$\wedge \backslash \wedge \backslash, \wedge \wedge \backslash \backslash$	2 ways
$n = 3$:	$\wedge \backslash \wedge \backslash \wedge \backslash, \wedge \wedge \backslash \wedge \backslash, \wedge \wedge \wedge \backslash \backslash, \wedge \wedge \wedge \wedge \backslash \backslash, \wedge \wedge \wedge \wedge \wedge \backslash \backslash$	5 ways

2.3.1 Into A Formula

Let's think of Dyck Paths with the line $y = x$ as the level they start on. Now, they are a series of steps from $(0, 0)$ to (n, n) that lie below the line.



We can approach this problem by thinking of every up step in the path as a 1 in the sequence, and every down step with -1 . There can never be more -1 s than 1s since we can't pass the line $y = x$. Thus the number of unique Dyck Paths becomes the number of sequences with non-negative partial sums that can be formed from n copies of 1s and n copies of -1 s. Look to section 2.4.1 for a proof to get a closed form for C_n

2.4 Parenthesis Problem

How many ways are there to count permutations of n pairs of parentheses such that each open parenthesis is followed by a closed one?

There is a recursive relationship here where we could add $()$ to the left, right, or around the whole thing. Going from $()$ to $()()$ could be seen as adding $()$ to the left or the right. Going from $()$ to $((()))$ would be adding parentheses around. Let's try small cases and list things out.

$n = 1$

- $()$

$n = 2$

- $()()$
- $(())$

$n = 3$

- $()()()$
- $(())()$
- $()(())$
- $(())()$
- $((()))$

With this, we can see that $C_1 = 1, C_2 = 2$, and $C_3 = 5$! This can be solved with coding! Here is a solution written in java

```

public List<String> generateParenthesis(int n) {
    List<String> list = new ArrayList<String>();
    backtrack(list, "", 0, 0, n);
    return list;
}

public void backtrack(List<String> list, String str, int open, int close, int max){

    if(str.length() == max*2){
        list.add(str);
        return;
    }

    if(open < max)
        backtrack(list, str+"(", open+1, close, max);
    if(close < open)
        backtrack(list, str+")", open, close+1, max);
}

```

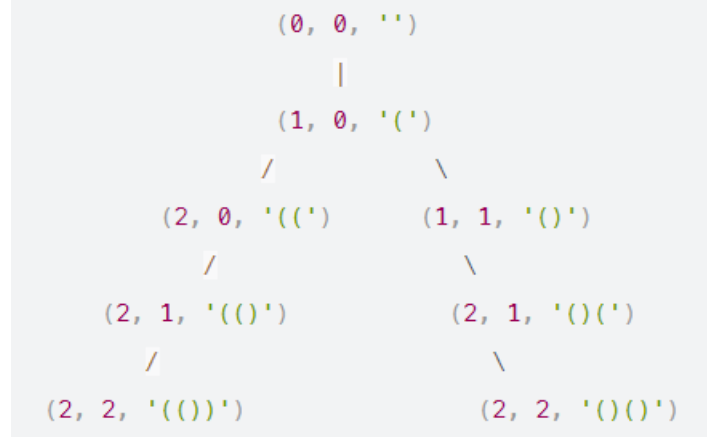
Let's analyze the lines under the line starting with **public void backtrack** in basic terms.

Let's look at **if (str.length() == max * 2)**. The word **str.length()** is actually a number, counting the amount of parenthesis we have added so far. **max** is a variable with the value of n where n is still the pairs of parentheses we want to find the permutations of. As we go through the rest of the code, keep in mind that $\text{max} = n$. With the **==**, we are checking if the number of total parentheses is equal to n . Since there are 2 parenthesis in each pair, we check if the amount of parentheses we have currently is equal to $n \cdot 2$, the final number of parentheses we want. If that is true, we stop adding more parenthesis.

Now let's observe **if (open < max)**. The word **open** is a variable counting the number of open parentheses we have, and we want n of them in total (n accounts for taking one from each of the n pairs we have). If we don't have enough of (, this means **open** is less than n , we want to proceed by adding another (, increasing the number of open parentheses we have, and branching off by going back to the first line under **public void backtrack**.

Taking a look at **if (close < open)**, the word **close** is a variable that counts the number of closed parentheses. Since we don't want to end up with situations like **) (** or **() (**, the number of closed parentheses at any given moment should be less than or equal to the number of open ones. If there are less closed than open, we can add another closed parenthesis.

In the end, we will end up accounting for all the permutations, since at every point, we choose to add (or) until we reach our goal. If we take a look at the picture below, each line in the form (open, closed, parenthesis added so far), we can see how the branching off doesn't affect each result. Finally, in the last line, we end up with the permutations we got above.



2.4.1 Into a Formula

It's nice that a computer can run it, but what if we wanted to compute by hand? Let's approach finding a formula by looking at the parenthesis problem.

The total number of sequences would be $\binom{2n}{n} = \frac{2n!}{n!n!}$. There are $2n$ total parenthesis and we divide by $n!$ twice to account for the parenthesis being indistinguishable from one another. Let's represent the open parenthesis (with 1 and closed parenthesis) with -1 and each number in the sequence as a_i with the entire sequence being $a_1, a_2, a_3, \dots, a_{2n}$. Every partial sum must be non-negative, that is

$$a_1 + a_2 + a_3 + \dots + a_k \geq 0$$

and $k = 1, 2, 3, \dots, 2n$. This means at any point in the sequence when looking at the numbers before, going from left to right, the number of 1s is greater than or equal to the number of -1 s. This makes sense when we think about the original context of the parenthesis problem: there are never more closing parenthesis than open ones.

Let A_n be the number of acceptable sequences and U_n be unacceptable sequences.

$$A_n + U_n = \binom{2n}{n} = \frac{2n!}{n!n!}$$

Let's try to count U_n by finding the first part the sequence breaks, that is to find the smallest k such that $a_1 + a_2 + a_3 + \dots + a_k < 0$. Since we are looking for the smallest k , there is an equal amount of 1s and -1 s before a_k . If that wasn't true, there would be a number before k where the sequence breaks, and k wouldn't be minimal. This means that k must be an odd number and $a_1 + a_2 + a_3 + \dots + a_{k-1} = 0$ and $a_k = -1$.

In the sequence up to a_k , which is $a_1, a_2, a_3, \dots, a_k$, there are $\frac{k-1}{2}$ copies of $+1$ s and $\frac{k+1}{2}$ copies of -1 s. In the remaining numbers, $a_{k+1}, a_{k+2}, \dots, a_{2n}$, there are $(n - \frac{k-1}{2})$ copies of $+1$ s and $(n - \frac{k+1}{2})$ copies of -1 s.

Let's consider replacing a_i where $i = 1, 2, 3, \dots, k$ with $-a_i$. This will make the sum of the entire sequence equal to 2: since the partial sum up to a_k was -1 and the rest of the sequence contains an extra copy of 1. Through this process, we made the sum up to a_k equal to 1. Now, in the whole sequence, there are $n + 1$ copies of 1s and $n - 1$ copies of -1 s. The number of possible sequences made by rearranging these 1s and -1 s would be $\frac{2n!}{(n+1)!(n-1)!}$. This number is equal to U_n since we can transform any sequence of this form into an unacceptable sequence by finding the point where there are more 1s than -1 s and negating all the numbers up to and including that point.

Now that we have $U_n = \frac{2n!}{(n+1)!(n-1)!}$, we can find A_n by substitution.

$$\begin{aligned}
 A_n &= \frac{2n!}{n!n!} - U_n &&= \frac{2n!}{n!n!} - \frac{2n!}{(n+1)!(n-1)!} \\
 &= \frac{2n!}{n!(n-1)!} \left(\frac{1}{n} - \frac{1}{n+1} \right) \\
 &= \frac{2n!}{n!(n-1)!} \left(\frac{1}{n(n+1)} \right) \\
 &= \frac{2n!}{n!n!} \frac{1}{n+1} \\
 &= \frac{1}{n+1} \binom{2n}{n}
 \end{aligned}$$

Thus, we get a closed form for C_n !

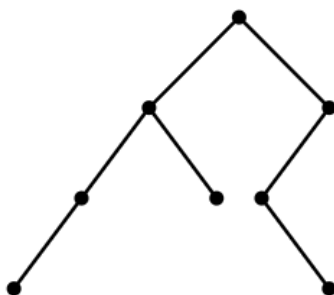
2.5 Ballot Problem

Suppose in an election, there are candidates A and B who each receive n votes. What is the number of sequences that A will never trail B when counting votes? For example, AAABBB does work, but AABBBBA doesn't work because after counting 5 votes, A has 2 votes which is less than the 3 votes B has.

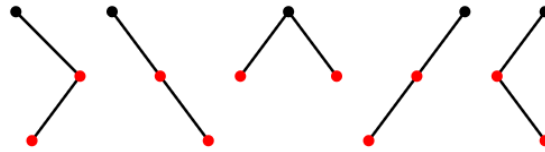
We can see a similarity between this and the parenthesis problem, especially when represented with 1s and -1 s. Since you can't have more closing parenthesis than open ones at any point in the sequence, the same idea persists in there never being more Bs than As. We can also see a parallel with the Dyck paths problem! Each 1 represents a step up and -1 represents a step down. Recall that we can't go lower than the starting height, so there can't be more steps down than steps up. Thus, the total number of sequences of all these situations would be C_n .

2.6 Rooted Binary Trees

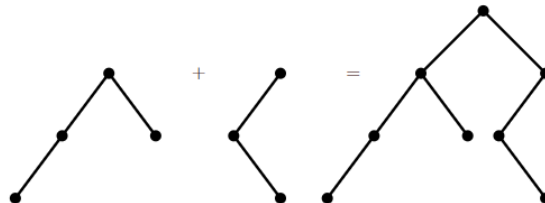
Rooted binary trees are a type of graph that appears in computer science. The tree is made up of vertices and edges where the vertices below a vertex connected by an edge are the children of that vertex. The root is the topmost vertex and a tree is binary when all vertices have either 0, 1, or 2 children. Below is an example of a rooted binary tree.



How many different rooted binary trees are there with n vertices (where the edges going in different directions count as different trees)? Try drawing out some small cases! There's 1 way for the tree to be empty, 1 way with one vertex, 2 ways with two, and 5 ways with three (the trees for 3 vertices are shown below).



These numbers look oddly familiar; they are Catalan numbers! In fact, we can think of making these trees by combining smaller ones.



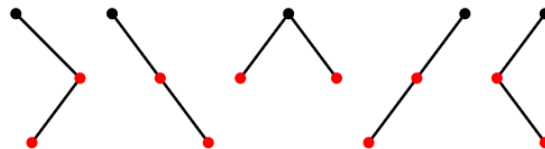
We start with n vertices, use 1 for a root vertex and its two children, and insert rooted binary trees on k and l vertices, with $k + l = n - 1$. We can write this as

$$C_n = \sum_{i=0}^{n-1} C_i C_{n-i-1}$$

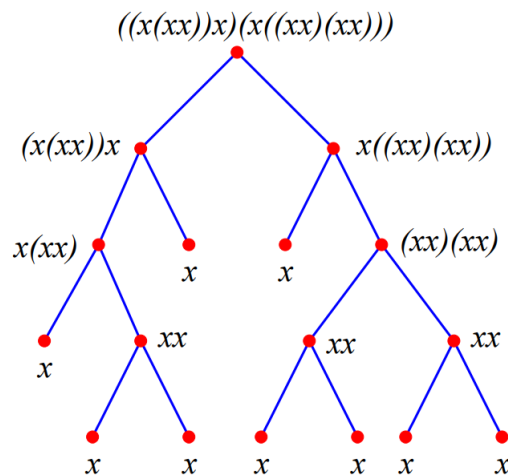
When we try to find C_3 with this formula, we get

$$C_3 = C_0 C_2 + C_1 C_1 + C_2 C_0 = 1 \cdot 2 + 1 \cdot 1 + 2 \cdot 1 = 5$$

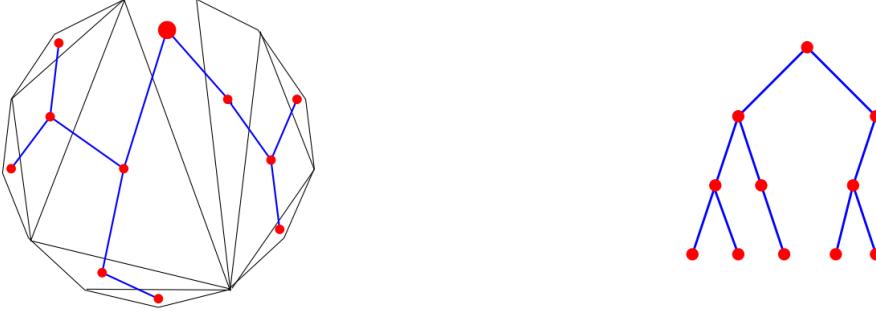
Now let's look at the trees with three vertices with this idea of combining trees. The two left trees have an empty left child from C_0 and a similar idea for the two right trees.



The parentheses problem can be mapped with a binary tree! Here's one that shows n pairs of parentheses.



Euler's polygon problem can similarly be mapped! Let each triangle represent a node and connect nodes based on what triangles share sides.



3 The n^{th} Catalan Number

As we have mentioned before, this is the explicit formula for C_n

$$C_n = \frac{1}{n+1} \binom{2n}{n} \quad (1)$$

Generating functions can be used to prove the explicit formula as well! Each term would be of the form $C_n x^n$ and let's define the function $C(x) = \sum_{n=0}^{\infty} C_n x^n$. We want the function to look like $1 + x + 2x^2 + 5x^3 + \dots$. After squaring both sides, we get

$$(C(x))^2 = \sum_{n=0}^{\infty} C_n x^n \cdot \sum_{n=0}^{\infty} C_n x^n$$

Since these are two infinite series, we can use properties of the Cauchy product

$$\left(\sum_{i=0}^{\infty} a_i \right) \cdot \left(\sum_{j=0}^{\infty} b_j \right) = \sum_{k=0}^{\infty} c_k \text{ where } c_k = \sum_{l=0}^k a_l b_{k-l}$$

Then the expression is rewritten as

$$\begin{aligned} (C(x))^2 &= \sum_{n=0}^{\infty} \left(\sum_{k=0}^n C_k C_{n-k} \right) x^n \\ &= \sum_{n=0}^{\infty} C_{n+1} x^n = \frac{C(x) - 1}{x} \end{aligned}$$

Let $y = C(x)$, and we can rewrite the equation as

$$\begin{aligned} y^2 &= \frac{y-1}{x} \\ xy^2 - y + 1 &= 0 \end{aligned}$$

We end up having a quadratic in y , so let's apply the quadratic formula and see what we get!

$$y = C(x) = \frac{1 \pm \sqrt{1 - 4x}}{2x}$$

Is the plus or minus sign the right one in this situation? Let's try expanding using Taylor Series.

$$\begin{aligned}\frac{1 + (1 - 2x - 2x^2 + \dots)}{2x} &= \frac{1}{x} - 1 - x + \dots \\ \frac{1 - (1 - 2x - 2x^2 + \dots)}{2x} &= 1 + x + \dots\end{aligned}$$

Clearly, we want the minus sign here, but we can also look at it with another method. We know that $C_0 = 1$, but $C(0)$ gives us an indeterminate form. However, we can take the limit as x approaches 0 from the positive side and see that the limit approaches 1.

$$\begin{aligned}\lim_{x \rightarrow 0^+} C(x) &= \lim_{x \rightarrow 0^+} \frac{1 - \sqrt{1 - 4x}}{2x} \\ &= \lim_{x \rightarrow 0^+} \frac{2(1 - 4x)^{-1/2}}{2} \text{ by l'Hôpital's rule} \\ &= 1\end{aligned}$$

We can use Newton's Binomial Theorem which says for any real number r that is not a non-negative integer when $-1 < x < 1$

$$(x + 1)^r = \sum_{i=0}^{\infty} \binom{r}{i} x^i$$

Now we can rewrite $\sqrt{1 - 4x}$ since $-1 < -4x < 1$ in order for the square root to be a real number.

$$\begin{aligned}\sqrt{1 - 4x} &= (1 + (-4x))^{1/2} = \sum_{n=0}^{\infty} \binom{1/2}{n} (-4x)^n \\ \frac{1 - \sqrt{1 - 4x}}{2x} &= \frac{1}{2x} \left(1 - \sum_{n=0}^{\infty} \binom{1/2}{n} (-4x)^n \right)\end{aligned}$$

Let's take a look at the coefficients of x^n in the sum

$$\begin{aligned}& \frac{\frac{1}{2}(\frac{1}{2} - 1) \cdots (\frac{1}{2} - n + 1)}{n!} (-4)^n \\ &= \frac{1(1 - 2) \cdots (1 - 2n + 2)}{n!} (-1)^n 2^n \\ &= \frac{(-1)(-3) \cdots (-(2n - 3))}{(n!)^2} (-1)^n 2^n (n)(n - 1) \cdots (1) \\ &= \frac{(-1)(-3) \cdots (-(2n - 3))}{(n!)^2} (-1)^n (2n)(2n - 2) \cdots (2) \\ &= -\frac{(2n)!}{(n!)^2 (2n - 1)} \\ &= -\binom{2n}{n} \frac{1}{2n - 1}\end{aligned}$$

Plugging this back into $C(x)$, we get

$$\begin{aligned}
C(x) &= \frac{1}{2x} \left(1 + \sum_{n=0}^{\infty} \binom{2n}{n} \frac{1}{2n-1} x^n \right) \\
&= \frac{1}{2x} \left(1 + (-1) + \sum_{n=1}^{\infty} \binom{2n}{n} \frac{1}{2n-1} x^n \right) \\
&= \frac{1}{2} \sum_{n=1}^{\infty} \binom{2n}{n} \frac{1}{2n-1} x^{n-1} \\
&= \frac{1}{2} \sum_{n=0}^{\infty} \binom{2(n+1)}{n+1} \frac{1}{2n-1} x^n \\
&= \sum_{n=0}^{\infty} \frac{2(n+1)(2n+1)}{2(n+1)^2} \binom{2n}{n} \frac{1}{2n+1} x^n \\
&= \sum_{n=0}^{\infty} \frac{1}{n+1} \binom{2n}{n} x^n
\end{aligned}$$

Finally, from this form, we can see that the coefficients of x^n , which are also C_n are

$$C_n = \frac{1}{n+1} \binom{2n}{n}$$

Alternatively, we can use the Generalized Binomial Theorem to see

$$\begin{aligned}
\sqrt{1-4x} &= \binom{1/2}{0} + \binom{1/2}{1}(-4x) + \binom{1/2}{2}(-4x)^2 + \dots + \binom{1/2}{k}(-4x)^k + \dots \\
\binom{1/2}{k} &= \frac{(1/2)(-1/2)(-3/2)\dots(1/2-k+1)}{k!} = \frac{(-1)^{k-1}1 \cdot 3 \cdot 5 \cdot (2k-3)}{2^k k!}
\end{aligned}$$

This means that the coefficient of x^k in $\sqrt{1-4x}$ is in the form

$$\frac{(-1)^{k-1}1 \cdot 3 \cdot 5 \cdot (2k-3)}{2^k k!} (-4)^k = \frac{(-1)1 \cdot 3 \cdot 5 \cdot (2k-3)2^k}{k!}$$

Let's use the substitution, $2^k k! = 2^k (1 \cdot 2 \cdot 3 \cdot \dots \cdot k) = 2 \cdot 4 \cdot 6 \cdot \dots \cdot k$, and multiply the number and denominator of the fraction on the left by $k!$

$$\begin{aligned}
\frac{(-1)1 \cdot 3 \cdot 5 \cdot (2k-3)2^k}{k!} &= \frac{(-1)1 \cdot 3 \cdot 5 \cdot (2k-3)2 \cdot 4 \cdot 6 \cdot \dots \cdot k}{k!k!} \\
&= \frac{(-1)(2k-2)!2k}{k!k!} \\
&= \frac{(-1)(2k)!}{(2k-1)k!k!} \\
&= \frac{-1}{2k-1} \binom{2k}{k}
\end{aligned}$$

Finally, we get that

$$\sqrt{1-4x} = -\sum_{k=0}^{\infty} \frac{1}{2k-1} \binom{2k}{k} x^k$$

$$C(x) = \frac{1 - \sqrt{1-4x}}{2x} = \frac{1 - \sum_{k=0}^{\infty} \frac{1}{2k-1} \binom{2k}{k} x^k}{2x}$$

From this form of $C(x)$, we can see that the coefficient of x^n in $C(x)$ is half of the coefficient of x^{n+1} in $\sqrt{1-4x}$ because of the $2x$ in the denominator.

$$\begin{aligned} \frac{1}{2} \binom{2(n+1)}{n+1} \frac{1}{2n+1} &= \frac{(2n+2)!}{2(n+1)!(n+1)!(2n+1)} \\ &= \frac{1}{2} \cdot \frac{2n+2}{n+1} \cdot \frac{(2n)!}{(n+1)!n!} \cdot \frac{2n+1}{2n+1} \\ &= \frac{1}{n+1} \frac{2n!}{n!n!} \end{aligned}$$

Now, we can use these coefficients to get the closed form!

$$C_n = \frac{1}{n+1} \frac{2n!}{n!n!} = \frac{1}{n+1} \binom{2n}{n}$$

4 Applications in the Real World

In addition to all the problems mentioned, Catalan numbers are seen in the fields of medicine, cryptography, geometric modeling, and more. Catalan numbers are used frequently to generate pseudo-random numbers. In cryptography, Catalan numbers are used to generate binary sequences which in turn produce keys. In steganography, concealing information, requires a set of constants which are generated with Catalan numbers. There are also applications in biometric identification, through triangulation and face recognition. There are still many problems that weren't mentioned, so feel free to explore more about this topic and do a deeper dive into its applications!

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